

2022

Q Distinguish between design synthesis and design analysis :-

B) ① Distinguish between design synthesis & design analysis.

→ a) Design Synthesis:

- ① It is a phase where designer generate & develop creative ideas & concepts to address specific design problem.
- ② It comes with innovative solutions to meet a set of requirement.
- ③ During design synthesis ideate, sketch & create prototypes to explore various design possibilities.
- ④ It's primary outcome is a collection of design concepts or proposals that represent potential solutions to the problem at hand.

b) Design Analysis:

- ① It is a phase where designers evaluate & assess the propose design concepts to ensure they meet the necessary criteria, standards & performance requirements.
- ② It involves a systematic examination of the feasibility & viability of the design ideas.
- ③ It analyze how well the design concepts align with the project goals & constraints.
- ④ The primary outcome of design analysis is a set of findings & recommendations.

Q. What is standardization? Explain its

2022 importance in design.

② What is standardization? Explain its importance in design.

→ Standardization refers to the process of developing and implementing consistent & uniform practices, specifications or guidelines for various processes, products or services across different industries & organizations. Standardization can apply to various fields, including manufacturing, technology, health-care & more.

Importance:

- 1) Quality Assurance: Standardization helps to ensure the quality & reliability of products & services.
- 2) Interoperability: In technology & various industries standardization is crucial for enabling interoperability.
- 3) Safety: Standardization also includes safety requirements & regulations.
- 4) Cost Reduction: Standardization can lead to cost savings by streamlining processes, reducing waste & minimizing the need for custom solutions.
- 5) Consumer Confidence: Consumers benefit from standardization by knowing that products & services meet certain quality & safety criteria.

Q. Write a short note on:-

2022

i) Aesthetic considerations in design.

ii) Factor of safety and its selection.

c) Write short note on :

1) Aesthetic consideration:

Aesthetic consideration is defined as a set of principles of appreciation of beauty. It deals with the appearance of the product. Appearance is an outward expression of quality of the product & is the first communication of the product with user. These consideration are highly influenced by individual preference, cultural norms & historical contexts.

2) Factor of Safety & its selection:

It is a critical engineering concept used to ensure the safety & reliability structures, components & systems. It represent the ratio of maximum allowable stress or load material can withstand actual applied stress or load.

The selection of Factor of Safety includes:

- ① Purpose of Factor of Safety
- ② Determining factor of safety
- ③ Factors influencing FOS selection.
- ④ Codes & standards
- ⑤ Optimization.

Q. Define Machine Design. Explain various design considerations.

[2019]

→ Defⁿ: Machine design is a specialized field of engineering that applies principles of physics, mechanics, and materials Science to create and develop machines and mechanical systems.

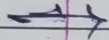
OR

Machine design is the processes of creating machines that do things for us. It involves understanding what the machine needs to do, designing its parts and how they work together, and building it.

* General Design Considerations :-

- 1.) Type of load and stresses caused due to type of load
- 2.) Motion of parts or kinematics
- 3.) Material selection
- 4.) Form and size of part
- 5.) Frictional resistance and Lubrication
- 6.) Economical and Convenient Features
- 7.) Use of standard part to Achieve Interchangeability.
- 8.) Availability of Workshop facilities
- 9.) No. of parts Manufactured
- 10.) Cost of Construction of the article (Manufacturing).
- 11.) Assembly of parts.

Q. Explain Aesthetic consideration in design
[2019] with sketches.



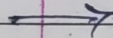
* Aesthetic Considerations:-

Aesthetics is defined as the set of principles of appreciation of beauty. It deals with the appearance of the product. Appearance is an outward expression of quality of the product and is the first communication of the product with user.

The following guidelines maybe used in Aesthetic design. (Design for appearance).

- 1.) The Appearance should contribute to the performance of the product.
- 2.) The appearance should reflect the function of the product.
- 3.) The appearance should reflect quality of the product.
- 4.) The appearance should not be at too much of extra cost unless it is a prime requirement.
- 5.) The appearance should be suitable to the environment in which the product is used.

Q. Explain ergonomics consideration in design
of control & display.



* ~~Def~~ Ergonomic Considerations:-

Ergonomics is defined as the scientific study of the man-machine working environment relationship and the application of Anatomical, Psychological ^{and Physiological} principles to solve the problem arising from this relationship. The final objective of ergonomics is to make the machine fit for user rather than to make user adapt himself.

or herself to the machine. It aims at decreasing physical and mental stress to the user.

* Ergonomic Considerations in Design of Display :-

1. The scale should be clear and crystal
2. Size of nos or letters on the scale should be taken appropriate.
3. Pointer should have knife edge with a mirror in a dial to minimize parallax error while taking reading
4. The scale should be divided in Linear progression such as 0, 10, 20, 30 & not as 0, 5, 25, 45...
5. The numbering should be in clockwise direction on a circular scale from left to right on horizontal scale and from bottom to top on a vertical scale.
6. The no. of sub-divisions should be as less as possible.

* Ergonomic Considerations in Design of Control :-

1. Controlled devices should be logically positioned and easily accessible.
2. Control operation should involve minimum and smooth movement.

3.7 The control operation should consume minimum energy

4.7 Proper colour should be used for control devices and backgrounds so as to give required psychological effect.

5.7 The shape and size of the control device should be such that the user is encouraged to handle it in such a way that to exert required force but not excessive force damaging control device or the machine.